
Part I: GRPO Finetuning of gemma-3-1b-it on GSM8K

Zixuan Wang (zw499)

Team **chmawa**: sm3035 · bc654 · zw499

I.1 Reproducing the baseline — the run & patches

(i) **The run.** Baseline reproduced as run **R0** ([8c2785ut](#)): the upstream **Tunix / JAX** GRPO pipeline (LoRA on frozen gemma-3-1b-it), commit borisbolliet/tpu-2026@77c5a67, run with the `config.py` defaults — KL-penalty weight $\beta=0.08$, group size $K=2$ (responses sampled per prompt), the full **3364** GRPO steps (one epoch), **~5 h 05 m** wall-clock on v6e-1.

Baseline patches (fixes for what did not run as-shipped; full diff in `tpu-2026_our_changes.patch`):

- TFDS's gsm8k builder crashes under protobuf 7.x once JAX imports — we evaluate the *identical* GSM8K test via an HF datasets fallback (seed 42);
- raised checkpoint retention (`SAVE_INTERVAL_STEPS` 500→100, `MAX_TO_KEEP` 4→40) so the best-val checkpoint survives selection (persistence, not training).

I.1 Reproducing the baseline — accuracy & training curves

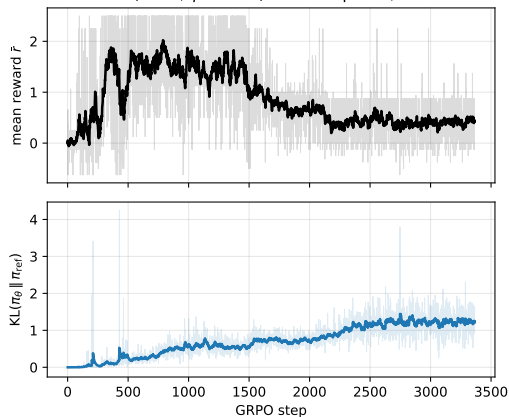
(ii) **Held-out accuracy.** GSM8K *test* (from HF), seed 42, greedy / exact-match, reported as ($n=64$ / $n=1319$) — $n=64$ is the `config.py` default eval, $n=1319$ the full test:

Model / checkpoint	$n=64$	$n=1319$
Base gemma-3-1b-it	45.3	47.4
R0 best-val (step 700)	42.2	46.7
R0 final (step 3364)	0.0	6.2

best-val = the saved checkpoint *nearest the peak held-out validation reward* (not accuracy).

(iii) Training curves (R0)

R0 baseline ($K=2$, $\beta=0.08$): reward peaks, then declines



I.2 Understanding the baseline

(a) The three policies (all share the frozen W_0):

- π_θ — LoRA-adapted policy; the **only** trainable policy ($\Delta W = \textcolor{red}{BA}$)
- π_{old} — the rollout policy; with $\mu=\text{NUM_ITERATIONS}=1$ (one gradient step per rollout batch), $\pi_{\text{old}}=\pi_\theta$ at the update
- π_{ref} — frozen base Gemma

(b) Group size & advantage. $K=\text{NUM_GENERATIONS}=2$; Tunix 'grpo' is the **standard** group-mean, std-normalised advantage over all K (incl. i) — *not* leave-one-out:

$$\hat{A}_i = \frac{r_i - \bar{r}}{\sigma_r}, \quad \hat{g} = \frac{1}{K} \sum_{i=1}^K h_i \hat{A}_i, \quad h_i = \nabla_\theta \log \pi_\theta(a_i | q).$$

I.2 Understanding the baseline

(c) **Reward = sum of four:**

- **Shaping:** `match_format_exactly`, `match_format_approximately` (the `<reasoning>/<answer>` template)
- **Correctness:** `check_answer` (gold match, only if the template parses) + `check_numbers` (any number after `<answer>`)

With the correctness reward *alone*, σ_r vanishes on most groups — not because the model is rarely right (base = 47%) but because the reward is **coarse**: at $K=2$ the two samples usually *tie*, so

$$\hat{A}_i = \frac{r_i - \bar{r}}{\sigma_r} = \frac{0}{\sigma_r + 10^{-6}} = 0.$$

The dense **shaping** terms keep $\sigma_r > 0$ and push toward parseable outputs (the precondition for `check_answer` to vary), although they also invite reward-hacking.

I.2 Understanding the baseline

(d) The clipped surrogate. The PPO-style clip lives in Tunix's GRPOLearner (`tunix.rl`); the clip range is $\epsilon=\text{EPSILON}=0.2$ in `config.py`. **Two deviations** from the lecture form:

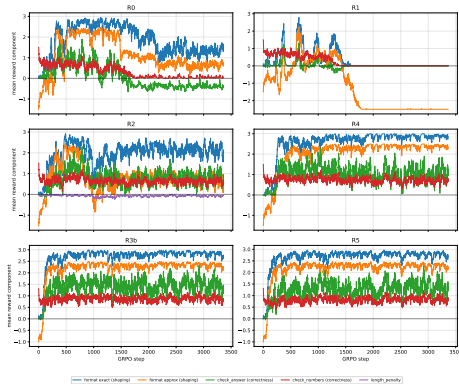
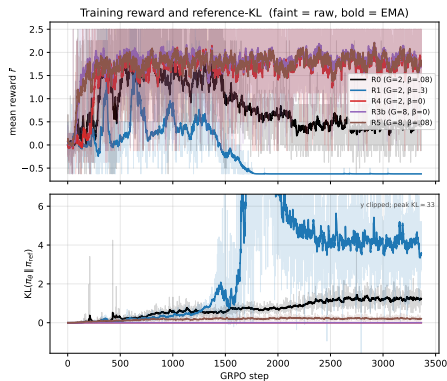
(i) Token- vs. sequence-level. The lectures clip a **per-response** ratio (one clip per whole sequence); Tunix clips a **per-token** ratio, token-by-token (a token-level objective):

$$\rho_i = \frac{\pi_{\theta}(a_i \mid q)}{\pi_{\text{old}}(a_i \mid q)} \quad \text{vs.} \quad \rho_{i,t} = \frac{\pi_{\theta}(a_{i,t} \mid q, a_{i,<t})}{\pi_{\text{old}}(a_{i,t} \mid q, a_{i,<t})}.$$

(ii) The clip is **inert in this baseline.** With $\mu=\text{NUM_ITERATIONS}=1$ (one gradient step per rollout batch) $\pi_{\text{old}}=\pi_{\theta}$, so every $\rho \equiv 1$ and **nothing is ever clipped** — the surrogate collapses to the plain policy gradient and ϵ is irrelevant. Not a *contradiction* (the lecture form also reduces to this at $\mu=1$), but the baseline's “clipped” objective does no clipping until $\mu > 1$.

I.3 Improving the baseline

Motivation. From the R0 training logs, we empirically found that the baseline **R0** **over-optimises** — training reward peaks then declines, and its reference-KL climbs **above 1** (*left*); the reward decomposition (*right*) reveals the mechanism — **R0's correctness** (check_answer) decays while format-shaping stays pinned high: **reward-hacking**.



I.3 Improving the baseline

Why R0 fails. The group-mean gradient has variance

$$\text{Var}(\hat{g}) = \frac{1}{K^2} \sum_i (h_i - \bar{h})(h_i - \bar{h})^\top = O(1/K),$$

maximal at the baseline's $K=2$ — the noisiest updates, which let the policy chase the dense **shaping** reward (reward-hacking) while true correctness decays.

Two predictions:

- (1) Variance.** Raising K shrinks $\text{Var}(\hat{g}) \propto 1/K \Rightarrow$ smoother updates, **milder over-optimisation**.
- (2) KL leash.** The reference-KL already climbs well above 1 (previous slide), so a *larger* β (tighter KL penalty) might rein in the reward-hacking.

Test: a controlled $\beta \times K$ 2×2 — **R4**(2, 0), **R0**(2, .08), **R3b**(8, 0), **R5**(8, .08) — plus **R1**(2, .3); **5 runs**, only the named knob changes.

I.3 Improving the baseline

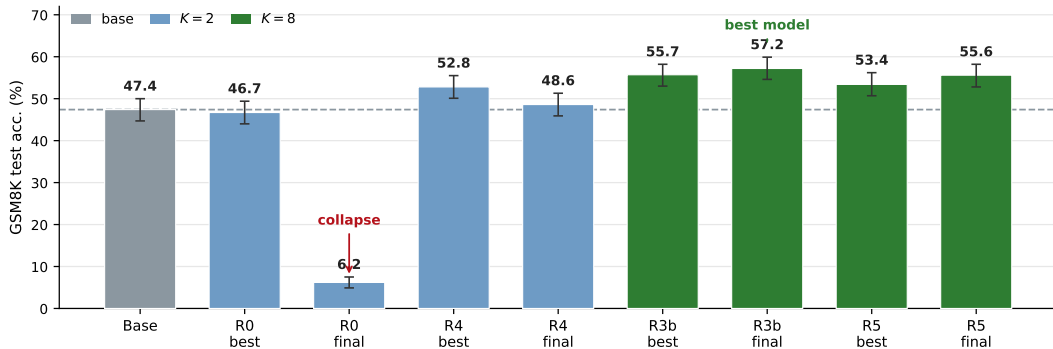
Held-out accuracy — full GSM8K test ($n=1319$, seed 42, greedy/exact); $\pm 95\%$ bootstrap CI and paired Δ vs. base over shared prompts. *: paired 95% CI excludes 0; †: collapse.

Model	(K, β) , step	Acc. (%)	95% CI	Δ base
Base gemma-3-1b-it	—	47.4	44.7–50.0	—
R0 (baseline)	(2, .08) best 700	46.7	44.0–49.4	−0.7 n.s.
R0	(2, .08) final 3364	6.2	4.9–7.5	−41.2 [†]
R4	(2, 0) best 1300	52.8	50.1–55.5	+5.4*
R4	(2, 0) final 3364	48.6	45.9–51.3	+1.2 n.s.
R3b	(8, 0) best 1000	55.7	53.0–58.2	+8.3*
R3b	(8, 0) final 3364	57.2	54.6–59.9	+9.9*
R5	(8, .08) best 1300	53.4	50.7–56.2	+6.1*
R5	(8, .08) final 3364	55.6	52.8–58.2	+8.2*

The β -axis extension **R1** (2, .3) fell **below base** (best 37.2 / final 0.0) — a clean negative result.

I.3 Improving the baseline

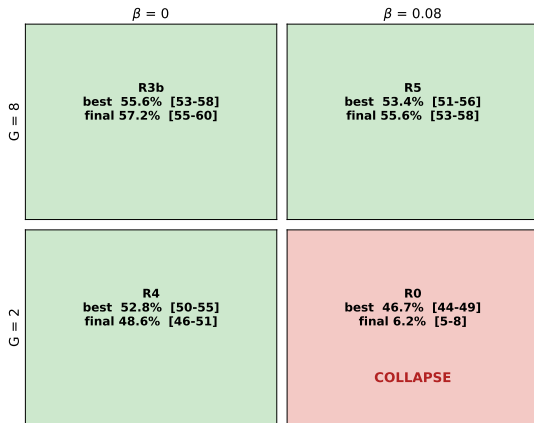
Held-out accuracy per checkpoint — the previous table as a bar chart (exact match, $\pm 95\%$ bootstrap CI):



R3b ($K=8$) tops out at **57.2%** (+9.9 pp over base); both $K=8$ runs beat base at best *and* final, while **R0**'s final checkpoint **collapses** to 6.2%.

I.3 Improving the baseline

Held-out GSM8K accuracy: group size $G \times$ KL weight β ($n=1319$)
collapse only at $G=2, \beta>0$

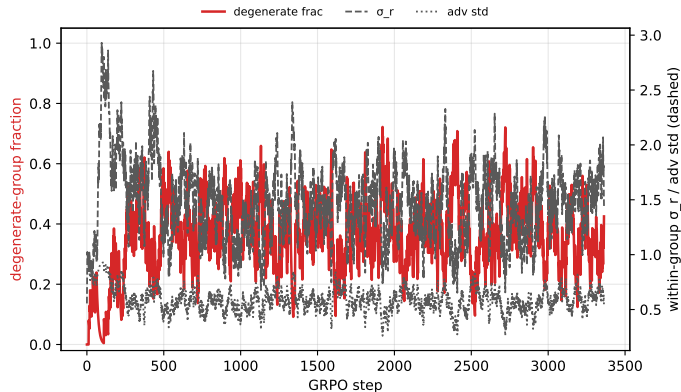


Prediction (1) holds. $K=8$ wins: **R3b** best at **57.2%** (+9.9 pp); both $K=8$ runs beat base and each beats its matched $K=2$ run. $K=2$ **peaks then degrades**, $K=8$ **improves monotonically** — over-optimisation is a $K=2$ phenomenon.

Prediction (2) contradicted. The tighter KL leash did *not* help: at $K=2$, $\beta=0$ beats $\beta=.08$ (R0 collapses) and $\beta=.3$ collapses *faster*; at $K=8$ the KL weight is irrelevant (R3b vs. R5: +1.7 pp, n.s.).

Cure: variance reduction (group size), *not* regularisation — echoing **DAPO**.

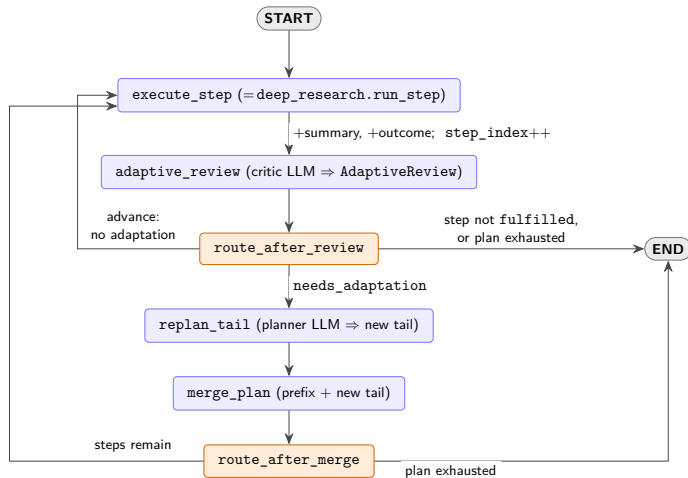
I.3 Improving the baseline



The cost of $K=8$: group degeneracy.

- About a **third of groups are degenerate** — all K rewards equal $\Rightarrow$ **zero advantage** (wasted samples; the dead-group / effective-sample-size failure mode).
- Yet within-group σ_r **holds** ≈ 1.5 — steady, *not* $\rightarrow 0$.
- Net effect still **+8–10 pp**.

Part II Adaptive planning in cmbagent_lg



Walkthrough.

- **Triggers a revision:** `adaptive_review` (critic LLM) sets `needs_adaptation`; the router replans *iff* the step was fulfilled, the plan is *not* exhausted, *and* adaptation is needed.
- **What may change:** *only* the **tail** (unrun steps); `merge_plan` = prefix + new tail.
- **Held fixed:** the completed prefix, its summaries / outcomes, and `step_index`.
- **Terminates:** when a step is not fulfilled or the plan is exhausted.

Part II Problems → proposed fixes

Problem	Fix
P1 Termination not <i>guaranteed</i> — the replanned tail can grow unboundedly; no monotone-decreasing quantity; <code>recursion_limit</code> <i>raises</i> , not finishes.	→ Thread a revision budget / max-steps ; routers finish gracefully; reject over-long tails. (<i>A hard cap can truncate a genuine expansion.</i>)
P2 Adaptation only on <i>success</i> — a <code>fulfilled=False</code> step routes straight to END; no retry or route-around.	→ Add a recovery / repair path on failure with a per-step retry budget. (<i>Needs a recoverable-vs-fatal flag.</i>)
P3 Replanned tail <i>bypasses</i> the reviewer that gates the initial plan — <code>merge_plan</code> splices it unchecked.	→ Route <code>new_tail</code> through the existing plan_reviewer for one critique round. (<i>One extra reviewer call.</i>)
P4 Partial-state feedback at unconditional cost — sees only the <i>last</i> step (Markov); fires every step.	→ Full-history, gated review — cheap model / skip when no downstream impact. (<i>Gate conservatively.</i>)